Optimizer Explained

What the RFR27 chassis optimizer does and how it works, with the math in plain words · October 2026

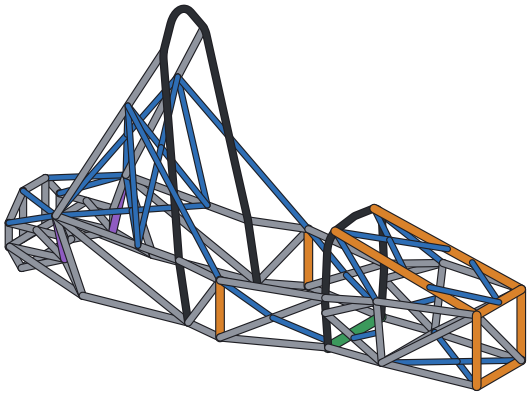
What it does

The optimizer designs the steel tube frame (the chassis) of the RFR27 Formula SAE car. Some things are fixed: the two roll hoops, the front and rear bulkheads, the 16 suspension pickup points, the driver and the space other parts need. The optimizer decides the rest: which tubes to add, where some joints sit and what size each tube is.

Its goal is simple: **the frame that twists the least for every pound of steel**, under a 75.5 lb weight cap, while following the FSAE rules and staying buildable. A fast computer model lets it try many thousands of designs. The best one is then built and tested in SolidWorks.

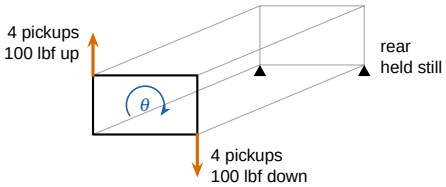
	TR (lb ft/deg)	Weight (lb)	TR per lb
Optimizer's design 013b	**2795**	**75.37**	**37.09**
Team Rev 8 2027	2150	75.57	28.45

013b: this pipeline's SolidWorks study. Rev 8: the team's TR tracker. That is about 30 % more stiffness per pound at the same weight.



Design 013b, drawn from its recipe. Black: hoops. Grey: 1 in tubes. Blue: 0.75 in. Orange: 1.25 in.

1 The score: how stiff is it in twist?



This is the team's torsional-rigidity (TR) test. Hold the 8 rear suspension points still. Push the 4 front-left points up and the 4 front-right points down, 100 lbf each. Then measure how far the front of the frame turns (RX, at the middle of the two front lower bars):

$$\theta = \tfrac{1}{2}(RX_1 + RX_2), \qquad \mathrm{TR} = \frac{545\ \mathrm{lb\,ft}}{\theta}.$$

TR is the torque it takes to twist the frame by one degree (545 lb ft is the tracker's torque for these forces). The score is TR per pound, TR/W, where W is the frame's weight: the higher it is, the more stiffness comes from every pound of steel.

2 The fast model: every tube is a set of springs

A SolidWorks study takes minutes, far too slow for thousands of designs, so the optimizer uses its own model of the frame, which solves in milliseconds. Each tube is a 3D beam. A round tube with outside diameter D, inside diameter d and length L has area A (how much steel) and section constants I and J (how well it resists bending and twisting). Its stiffness k_e is the sum of six simple patterns b_t (stretch, twist and four bending patterns), each with its own spring number s_t:

$$A = \tfrac{\pi}{4}(D^2 - d^2), \qquad I = \tfrac{\pi}{64}(D^4 - d^4), \qquad J = 2I, \qquad k_e = \sum_{t=1}^{6} s_t\, b_t b_t^{\mathrm{T}}.$$

The spring numbers are EA/L (stretch), GJ/L (twist) and, for bending in each of two directions, $3EI/((1+\phi)L)$ and EI/L, with E = 205 GPa and G = 80 GPa for 4130 steel and ϕ a small correction for shear. Changing a tube's size changes only its six numbers s_t, which is what makes the search fast. Adding every tube's springs together at the joints gives the whole frame:

$$K\,u = f, \qquad \theta = c^{\mathrm{T}} u, \qquad W = \sum_i \rho\, A_i\,(L_i - 25.4\ \mathrm{mm}).$$

K is the frame's stiffness, f the test forces, u how far every joint moves and turns, and c picks out the two probe rotations. For the weight, ρ = 7850 kg/m^3 and each tube end sits about 12.7 mm inside the tube it is welded to. Tubes whose surfaces come within 0.05 in of each other are joined, as in SolidWorks. This model reads about 1 % below SolidWorks.

3 The whole problem in one line

$$\text{maximize} \quad \frac{\mathrm{TR}(p,x)}{W(p,x)}$$

by choosing p: 9 joint positions, each inside the window the rules allow, plus 2 layout choices;
x: for every candidate tube, leave it out or use one of the sizes the supplier (VR3) stocks;

such that $W \le 75.5$ lb, the FSAE rule structure is there, every tube clears the driver and the keep-out zones, no two tubes cross, and tubes meet at 20° or more.

Two loops solve it: the inner loop picks the tubes for one set of joint positions, the outer loop moves the joints.

4 Inner loop: choosing the tubes

1. **Draw every option** (the "ground structure"): hundreds of candidate tubes between fixed points and points along other tubes, in every stocked size. Any that hit the driver or a keep-out zone are dropped; pairs that cross or meet at under 20° become conflicts.
2. **Find which tubes would help: the adjoint trick.** Re-solving the frame once per candidate would be slow. Instead, solve it once more with the probes as the load, $K\lambda = c$. Then, for every candidate at once,
$$\frac{\partial \theta}{\partial s_t} = -(b_t^{\mathrm{T}} u)(b_t^{\mathrm{T}} \lambda).$$
A tube helps when the twisting frame stretches or bends it ($b_t^{\mathrm{T}} u$) in a way the probes feel ($b_t^{\mathrm{T}} \lambda$).
3. **Check a change exactly: the Woodbury formula.** Adding, removing or resizing a tube changes only a few numbers in K (a small change UDU^{T}), so the new twist follows exactly from the old solution, in about a millisecond:
$$\theta_{\mathrm{new}} = \theta - (U^{\mathrm{T}}\lambda)^{\mathrm{T}}(I + DS)^{-1} D (U^{\mathrm{T}} u), \qquad S = U^{\mathrm{T}} K^{-1} U.$$
4. **Keep the best change** if TR per lb goes up. A short "tabu" list stops the search from undoing what it just did. Left and right tubes always change together, so the frame stays symmetric.
5. **Combine changes with a small integer program.** Each possible change k gets a yes/no variable x_k and its exact gain g_k. The solver (HiGHS, via SciPy) then picks
$$\text{maximize } \sum_k g_k x_k \quad \text{with at most one option per tube pair, no two crossing tubes, and } \sum_k x_k \le 6.$$
6. **Shake it up.** Random "kicks" and several restarts stop the search from settling on the first good answer.

Weight cap. While searching, each pound over 75.5 lb costs 2 points of TR per lb: merit = $\mathrm{TR}/W - 2\max(0, W - 75.5)$. At the end, the design is trimmed back under the cap.

5 Outer loop: moving the joints

Nine joint positions move inside the windows the rules allow: cockpit width and position, two side-member heights, where the main-hoop brace attaches and where it lands (3 coordinates), and a battery post, plus two layout choices. First the optimizer tries a spread of layouts covering every window evenly (Latin hypercube sampling). Then a compass search works around the best one: nudge each value up and down, keep any improvement, halve the step when nothing helps. Every try runs the whole inner loop.

6 Staying legal and buildable

- **Rules are checked from the geometry, never from tube names.** For example, the side impact structure (F.6.4) needs a Size B upper member whose top is 290–345 mm above the bottom of the lower member, Size B diagonals, and every bay between the hoops a triangle. The bulkhead support (F.6.2.3) and supports at tube bends (F.5.2.3) are checked the same way. A "free skeleton" mode may re-route these tubes while the checks pass; it gained only 0.7 %, as the packaging zones limit the design, not the rules.
- **Keep-out zones.** The driver and cockpit templates, the battery and its drop-out path, the tie rods, shocks, master switch and chain each get a convex zone. Several were estimated from photos of the team's previous car. Every tube's clearance to every zone is checked exactly along its whole length, using its real diameter.
- **Buildable.** Only stocked tube sizes, and every tube is cut to fit (coped) in an order the car could really be welded in (next section).

7 How the trimming order is chosen

It is chosen automatically, by a fixed rule rather than by optimization. The team procedure wants tubes trimmed in an order the car could be welded in, so **every tube is coped onto the tubes placed before it, never the other way round**:

1. **Build order:** the hoops first, then the rule structure in a set sequence (front bulkhead, rear bulkhead, main-hoop braces, side members, …), then the optimizer's added tubes, X-braces last.
2. **Who cuts whom:** a new tube is coped onto every earlier tube it touches (a shared joint, or its end lands on that tube). The earlier tube is never cut. Design 013b has 460 copes, 38 onto the hoops.
3. **Special cases:** X-brace halves are coped only onto the tube they cross; at a kink a tube is coped first onto the member it sits on.
4. **Trim as added:** SolidWorks adds a group of tubes, cuts it at once, saves and moves on. Cuts leave a 0.1 mm gap (0.05 mm under 1 in), so the simulation still joins every tube end.

Not checked: whether a welder's torch can reach every joint in this order.

8 Checking in SolidWorks

The best design is built automatically in SolidWorks as a weldment. SolidWorks checks the rule templates, driver, keep-out zones, tube overlaps and weight, then runs the team's TR test on the finished part; a result counts only if the solver's input checks out. The fast model searches, SolidWorks has the final say.

TR per lb	Fast model	SolidWorks
Design 011b	35.44	35.75
Design 013b	36.65	37.09